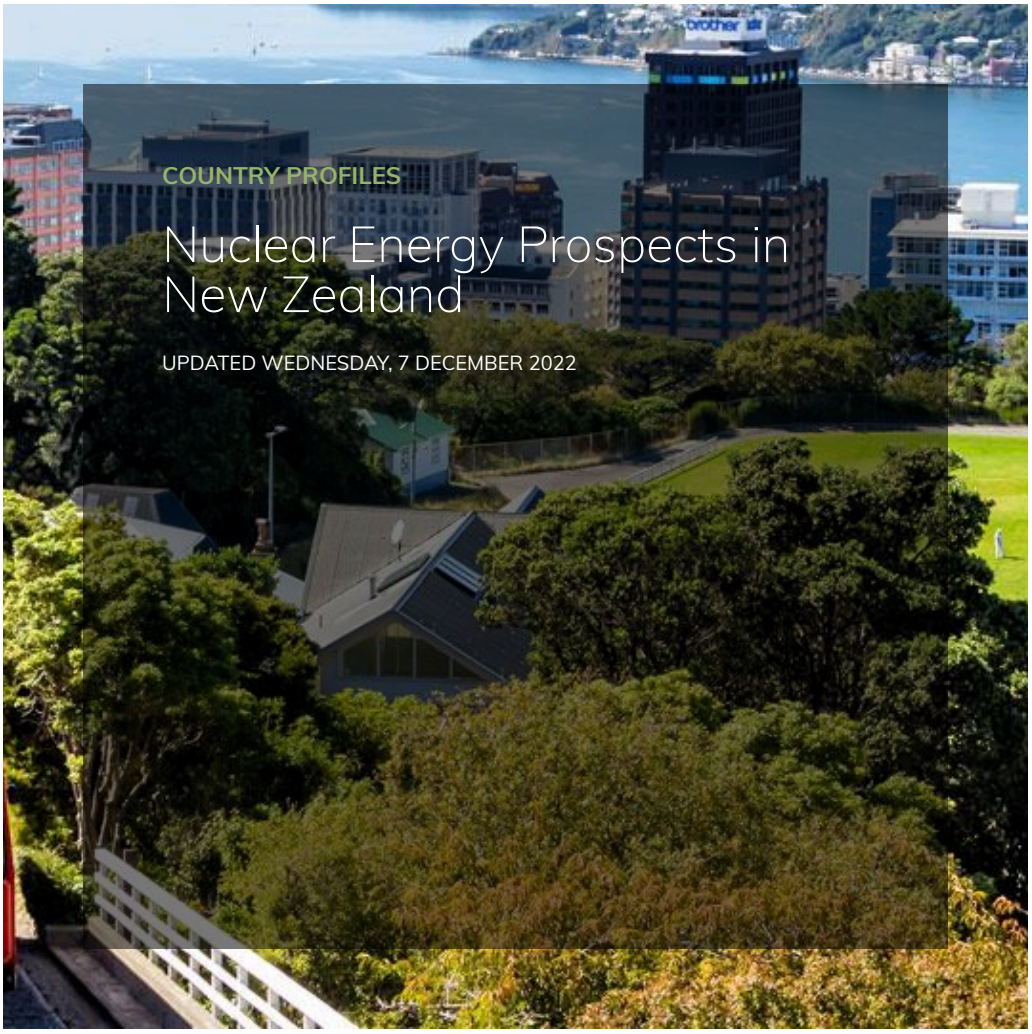

- New Zealand is one of the few developed countries not using electricity (indigenous or imported) from nuclear energy.
- As hydroelectric potential was progressively utilized, nuclear power featured in national power plans from 1969 to 1976, **but was not pursued.**

Electricity sector

Privacy - Terms

Total generation (in 2020): 45.2 TWh

Generation mix: hydro 24.2 TWh (54%); geothermal 8.1 TWh (18%); natural gas 6.3 TWh (14%); coal 2.4 TWh (5%); wind 2.3 TWh (5%); biofuels 1.6 TWh (4%); solar 0.2 TWh.

Import/export balance: no electricity imports or exports.

Total consumption: 39.2 TWh (2019 data)

Per capita consumption: 7900 kWh (2019 data).

Source: International Energy Agency and The World Bank. Data for the year 2020 unless specified.

New Zealand has depended primarily on hydroelectric power for its electricity for many years, but scope for expansion is limited and there has been no large-scale increase in hydro capacity since the Clyde Dam on the Clutha River was commissioned in the early 1990s[a]. As a result, growth in demand since 1990 has been mostly met by geothermal generation and gas-fired plants (at least until the 1000 MWe state-owned Huntly plant shifted to using coal for 80% of its energy[b]). Geothermal energy provides up to a fifth of the country's electricity generation, with most capacity situated in the Taupo Volcanic Zone on North Island. For the population of 5 million, average per capita consumption is about 7900 kWh per year, or considerably less if aluminium smelting is treated as largely an electricity export.[c]

Most of the demand for electricity comes from North Island, notably Auckland. The majority of hydroelectric generation comes from South Island, with large schemes built in the MacKenzie basin. Although there are also projects on North Island, by the Waikato river and the Central Plateau basin, which is also where the majority of New Zealand's geothermal is generated.

The HVDC Inter-Island link – commonly referred to as the Cook Strait cable – is a multiple 1040 MWe HVDC link connecting North and South islands, owned and operated by state-owned transmission company Transpower New Zealand.

Auckland's power supply is particularly vulnerable to power outages. In March and April 1998 the city suffered a five-week-long outage. More recently, both minor and major interruptions have occurred. New base-load capacity is required nationally.

Nuclear-related policy

In 1968 the national power plan first identified the likely need for nuclear power in New Zealand a decade or more ahead, since readily-developed hydro-electric sites had been utilized. Plans were made and a site at Oyster Point on the Kaipara harbour near Auckland was reserved for the first plant. Four 250 MWe reactors were envisaged, to supply 80% of Auckland's needs by 1990. But then the Maui gas field was discovered, along with coal reserves near Huntly, and the project was abandoned by 1972.

In 1976 the Royal Commission on Nuclear Power Generation in New Zealand was set up to inquire further into the question. Its 1978 report said that there was no immediate need for New Zealand to embark upon a nuclear power program, but suggested that early in the 21st Century "a significant nuclear programme should be economically possible".

In 1987 the New Zealand Nuclear Free Zone, Disarmament, and Arms Control Act was passed[e]. This was largely a symbolic statement of opposition to nuclear war and weapons testing, and it prevented the visits by nuclear-propelled or nuclear-armed vessels (primarily US ones). The New Zealand Nuclear Free Zone established under the Act does not ban land-based nuclear power plants.

Public opinion

In 2005 a survey of business leaders showed that 94% were concerned about future energy supply in New Zealand, and nearly two-thirds supported investigation of nuclear power.

The most recent survey of public opinion was in 2008. 19% of respondents to the survey chose nuclear as among the best options for New Zealand.

Uranium

No significant uranium deposits are known, but there is a proposal to mine phosphate from the sea bed on the Chatham Rise, extending 1000 km offshore east of the South Island. An average of 240 ppm uranium occurs in the phosphates, which would be dredged and processed onshore at the rate of about 1.5 million tonnes per year, hence potentially yielding up to 360 tU per year. Chatham Rock Phosphate is the project company.

However, in 2015 Chatham Rock Phosphate's proposition to mine phosphorite nodules from the crest of Chatham Rise was refused due to environmental concerns. The company plans to reapply for consent in the late 2020s.

Further information

Notes

a. The 432 MW Clyde Power Station on Lake Dunstan commenced operation in 1992 after overcoming much opposition to the development. It is one of two large dams on the Clutha River owned and operated by Contact Energy, and together they account for around 10% of the country's total electricity production. The Clyde Dam is the third largest dam in New Zealand. Contact Energy is the second largest generator in New Zealand and in 2008 accounted for 24% of the country's electricity generation. The company was split from the state-owned Electricity Corporation of New Zealand (ECNZ) in 1996 and is 51% owned by Australian company Origin Energy.

In late 2008, Contact Energy confirmed it was considering old plans for for a number of possible future hydro developments ranging from 86 MW and 350 MW on the Clutha River. These plans were originally developed by Contact's predecessor, ECNZ. Other hydro projects under consideration are: Meridian Energy's North Bank Tunnel Project on the Waitaki River in South Canterbury, which would have a capacity of around 200 MW; TrustPower's 72 MW Wairau scheme in Marlborough; and Meridian Energy's 65-85 MW Mokihinui station on the West Coast of the South Island. [Back]

b. The 1448 MW Huntly Power Station located 70 km south of Aukland is New Zealand's largest thermal power station. The station consists of: four 250 MWe units (units 1-4) that can burn coal or natural gas; a 400 MWe combined cycle gas turbine (unit 5); and a 48 MWe open cycle gas turbine (unit 6). The station is owned and operated by state-owned Genesis Energy. [Back]

c. Most of the output of the New Zealand's largest hydroelectric power station, Manapouri Power Station[d], is supplied to New Zealand Aluminium Smelters Limited (NZAS) located on Tiwai Point, near Invercargill. NZAS is a joint venture between Rio Tinto Alcan (79.36%) and Sumitomo Chemical Company (20.64%). Rio Tinto

Alcan is the successor company to Comalco, which was acquired by Rio Tinto in 2000. Comalco was renamed Rio Tinto Aluminium in late 2006 before being renamed Rio Tinto Alcan following Rio Tinto's acquisition of Alcan. The smelter, which is one of the world's largest, was opened in 1971. NZAS is the largest single consumer of electricity in New Zealand. [Back]

d. The underground Manapouri Power Station commenced operation in 1969. The station is owned and operated by state-owned Meridian Energy, which is the largest generator in New Zealand, accounting for 27% of production in 2008. Manapouri's design capacity was 700 MW, but could not be operated above 585 MW due to larger than expected friction between the water and the tailrace tunnel. Between 1997 and 2002, a second tunnel was constructed, allowing operation at maximum capacity. Between 2002 and 2008, the seven generating units were upgraded to 121.5 MW each, bringing the total installed capacity to 840 MW. However, it can only operate at less than 730 MW due to limits required by the resource consents. Meridian is seeking new resource consents, which would allow Manapouri to be operated at 830 MW. [Back]

e. In 1984, the new Labour government pledged to declare New Zealand nuclear free. In February 1985, New Zealand refused entry to the destroyer *USS Buchanan*, resulting in the USA suspending its obligations towards New Zealand under the Australia, New Zealand, United States Security (ANZUS) Treaty. Then, under the Nuclear Free Zone, Disarmament, and Arms Control Act 1987, the New Zealand government established the New Zealand Nuclear Free Zone. The text of the New Zealand Nuclear Free Zone, Disarmament, and Arms Control Act 1987 is on the Parliamentary Counsel Office website for New Zealand Legislation (www.legislation.govt.nz).

In May 2000, the Green Party introduced the New Zealand Nuclear Free Zone Extension Bill, which sought to extend the nuclear-free zone up to the 200-mile (370 km) exclusive economic zone (the Act covers up to 12 miles from the shore), but this was thrown out 108 to 7 two years later.

In July 2005, a bill to delete Section 11, which bans the entry of nuclear-propelled ships into New Zealand waters, was thrown out 107 to 9. [Back]

General sources

Nuclear Power Generation in New Zealand: Report of the Royal Commission of Inquiry, Royal Commission on Nuclear Power Generation in New Zealand, 1978
Energy strategies for New Zealand page on the Ministry of Business, Innovation & Employment website
New Zealand Emissions Trading Scheme page on the Ministry for the Environment website

Related information

Renewable Energy and Electricity